

IDX Supercomputer Dashboard

Complete Documentation Guide

Indonesian Stock Exchange Analytics Terminal

Version 3.0 — March 2026

This document provides a comprehensive explanation of every section, metric, and data point displayed on the IDX Supercomputer Dashboard. It is written for readers who may not have a background in stock trading or financial markets. Each term is defined in plain language so that anyone can understand the information presented on the platform.

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1. Overview

The IDX Supercomputer Dashboard is a web-based analytics terminal displaying data about stocks listed on the Indonesia Stock Exchange (IDX / Bursa Efek Indonesia). Inspired by Bloomberg Terminal, Stockbit, and Ajaib, it is organized into seven tabs: Overview, Rankings, Screener, Intelligence, Supply Chain, Informal Economy, and Stream.

Market data is sourced from official IDX Daily Stock Summary files (idxdata3.co.id) when available. When unavailable, the last official snapshot or a bootstrap dataset is used, clearly labelled.

1.1 Tech Stack

Component	Technology	Version
Language	JavaScript (ES2020+)	Node.js v24.14.1
Runtime	Node.js	v24.14.1
Web Framework	Express.js	v4.18.2
HTTP Client	Axios	v1.6.7
Task Scheduler	node-cron	v3.0.3
IDX Data Parser	dbffile (dBASE III/IV)	v1.12.0
ZIP Extraction	adm-zip	v0.5.16
Authentication	Firebase Auth (CDN)	v10.12.0
Database (Stream)	Firebase Firestore (CDN)	v10.12.0
Frontend	Vanilla HTML5, CSS3, JS	—
CSS Theme	Custom properties, dark mono	—
Font	Arial / Helvetica (system)	—
Deployment	Vercel (@vercel/node)	—
Inspiration	Bloomberg, Stockbit, Ajaib	—

1.2 Data Disclaimer

DISCLAIMER

The market data currently displayed on this dashboard is generated using deterministic algorithms and does not represent real-time or historically accurate IDX stock prices. This data is for demonstration and development purposes only. Actual IDX market data will be integrated once a subscription to official IDX data services or a verified third-party data provider has been established. All metrics, scores, and derived indicators are subject to change. Users should not make financial decisions based on the information currently displayed.

2. Key Concepts and Terminology

Essential terms specific to the Indonesian stock market:

Term	Definition
IHSG / JCI	Jakarta Composite Index. Tracks all IDX stocks.
Ticker	Four-letter code (e.g., BBCA = Bank Central Asia).
HAKA	Aggressive buying at ask price. High = strong buying.
HAKI	Aggressive selling at bid price. High = strong selling.
ARA	Max daily price increase (+20–+35%).
ARB	Max daily price decrease (typically " 7%).
Bandar	Large players whose activity moves prices.
Foreign Flow	Net buying/selling by international investors.
PER	Price-to-Earnings Ratio.
PBV	Price-to-Book Value.
ROE	Return on Equity.
RSI	Relative Strength Index (0–100).
Volume	Total shares traded.
Value	Total trade value, millions IDR.
Broker Summary	Largest brokerages per stock per day.

3. Dashboard Sections

3.1 Overview Tab

Landing page: six summary cards, IHSG panel with breadth bar and sector heatmap, top 15 movers, money flow table, ARA/ARB monitor, 12-item news feed.

3.2 Rankings Tab

Best/worst 20 stocks across five timeframes. Daily ranks by Day Trade Score; Yearly by Investment Score.

3.3 Screener Tab

Full table of 60 stocks: Ticker, Name, Open, High, Low, Close, Prev, Chg%, Volume, Value, Bid, Ask, Source.

3.4 Intelligence Tab

Unified News + Koneksi & Kultur view with six sub-tabs:

- News Feed: articles by timeframe with source, category, sentiment, impact, tickers.
- Political Risk: regulatory impacts, tenders with anomaly detection, connection heatmap.
- Social Sentiment: influencer tracker, retail chatter, conglomerate drama, Gojek/Grab proxy.
- Cultural Catalysts: seasonal events, viral/meme alerts, regional pride, campus hype.
- Conglomerates: ten families with succession risk, heir readiness, net worth.
- Wealth Transfer: generational preferences, family offices, succession scorecards.

3.5 Supply Chain Tab

- Commodity Dashboard: CPO, Nickel, Coal, Gold, Copper, Tin, Rubber, Natural Gas.
- Port & Logistics: utilization, congestion, wait times.
- Weather Impact: monsoon, El Niño/La Niña, volcanic alerts.
- China Dependency: export, import, investment exposure per stock.
- Inter-Island Disparity: Java vs outer islands.

3.6 Informal Economy Tab

- Warung Index: food stall/shop foot traffic.
- Ojek Demand: ride-hailing across six cities.
- Pasar Prices: staple goods as inflation signal.
- Remittance Flow: overseas worker remittances.

3.7 Stream Tab

Reddit-style discussion: posts with tags, upvote/downvote, comment threads. Requires auth.

4. Stock Detail Modal

Clicking any ticker opens a detailed popup:

Section	Content
Price Display	Current price, change, mini intraday chart.
OHLC	Open, High, Low, Close.
Source Reference	Link to official IDX file and trade date.
Trading Data	Volume, avg volume, ratio, value, bid/ask, spread.
HAKA/HAKI	Buy/sell pressure scores with visual bar.
ARA/ARB Levels	Daily limits, distance, alerts.
Technical	RSI (14), MACD, SMA 20/50/200.
Foreign Flow	Buy, sell, net, % of volume.
Bandar Activity	Score (0–100), accumulation signal.
Fundamentals	Market cap, PER, PBV, ROE, DER, div yield.
Multi-Timeframe	Daily/weekly/monthly/quarterly/yearly.
Composite Scores	Day Trade and Investment Score (0–100).
Top Brokers	Top 5 buy and 5 sell brokerages.

5. Authentication System

The Stream tab requires Firebase Authentication. Three methods:

Email and Password Registration

Users provide a username (3–24 chars), email, and password meeting:

- Minimum eight characters.
- At least one uppercase letter.
- At least one lowercase letter.
- At least one numeric digit.
- At least one special character (!, @, #, \$, etc.).

A real-time strength bar ranges from "Very weak" (red) to "Very strong" (cyan). Submit stays disabled until all rules pass and passwords match.

Email Verification

Firebase sends a verification link. Users must verify before posting. A banner with "Resend Email" and "I've Verified" buttons shows until complete.

Anonymous Sign-In

Browse and interact anonymously. Labelled "Anonymous" on all content.

6. Data Sources and References

Every API response includes a source field indicating data origin:

Source	Description
IDX Stock Summary	Official daily files from idxdata3.co.id. OHLC, volume, value, bid, ask for all listed stocks. Updated each trading day.
Reference URL	Direct link to downloaded archive (e.g., SS260317.zip) for verification.
Bootstrap Fallback	Generated data when official source is unreachable. Labelled "Fallback".
Cached Snapshot	Last official fetch retained when refreshes fail. Labelled "(cached)".
Firebase Firestore	Cloud NoSQL database for Stream posts, comments, votes, user data. Auth via Firebase Authentication.

API metadata fields:

- provider: name of the data source.
- fileName: specific file fetched.
- fileUrl: download URL for verification.
- fetchedAt: ISO 8601 timestamp.
- stale: boolean for cached/fallback data.

7. Glossary

Alphabetical quick-reference:

Term	Definition
Advancing	Price increased vs previous day.
ARA	Max daily price increase.
ARB	Max daily price decrease.
Ask	Lowest price a seller will accept.
Bandar Score	Large player accumulation (0–100).
Bid	Highest price a buyer will pay.
Breadth	Advancing vs declining ratio.
BUMN	State-owned enterprises.
Day Trade Score	Short-term attractiveness (0–100).
Declining	Price decreased vs previous day.
DER	Debt-to-Equity Ratio.
Div Yield	Annual dividend / stock price.
Foreign Net	Foreign buy " sell, millions IDR.
HAKA	Aggressive buying at ask price.
HAKI	Aggressive selling at bid price.
IHSG	Jakarta Composite Index.
Invest Score	Long-term attractiveness (0–100).
MACD	Moving Avg Convergence Divergence.
Market Cap	Price × shares outstanding.
PBV	Price-to-Book Value.
PER	Price-to-Earnings Ratio.
ROE	Return on Equity.
RSI	Relative Strength Index (0–100).
SMA	Simple Moving Average (N days).
Spread	Bid-ask gap in basis points.
Vol Ratio	Today vol / avg. >1 = above normal.

